

EDUCATION

National University of Singapore

Ph.D. in Mathematics, Advisor: Tan Minh Nguyen

Singapore

2022–Present

Vietnam National University – University of Science

B.A. in Mathematics, with Honors

Hanoi, Vietnam

2018–2022

RESEARCH INTERESTS

- **Symmetry in Parameter Space.** I study a central question: given a model f , how are parameter settings θ and $\bar{\theta}$ related when they represent the same function, $f_\theta \equiv f_{\bar{\theta}}$? Applications include understanding implicit bias through neural conservation laws, symmetry-aware optimization and quantization, equivariant metanetworks, and mode connectivity up to symmetry.
- **AI for Mathematical Reasoning.** I am also interested in using AI to tackle mathematical problems, particularly in commutative algebra and algebraic geometry, to better understand the capabilities and limitations of large language models in mathematical reasoning.
I share updates on this work at <https://vh-tran.github.io/ai-in-maths.html>.

PUBLICATIONS

[†] denotes first or co-first authorship.

Conference Publications

1. **Tree-Sliced Orlicz Integral Probability Metric**
Tuan A. Hoang[†], Trung-Khang Tran[†], **Viet-Hoang Tran**[†], Tan Minh Nguyen.
Advances in Neural Information Processing Systems (NeurIPS), 2026.
2. **Conformal Prediction for Distribution-to-Distribution Regression**
Trung-Khang Tran[†], Tuan A. Hoang[†], **Viet-Hoang Tran**[†], Tan Minh Nguyen.
Advances in Neural Information Processing Systems (NeurIPS), 2026.
3. **Conservation Laws for Modern Neural Architectures**
Viet-Hoang Tran[†], Vinh Khanh Bui[†], Tan Lai Ngoc, Nam Nguyen, Tuan Quang Dam, Tan Minh Nguyen.
International Conference on Machine Learning (ICML), 2026 (spotlight).
4. **Functional Equivalence in Attention: A Comprehensive Study with Applications to Linear Mode Connectivity**
Viet-Hoang Tran[†], Khanh Vinh Bui, Van-Hoan Trinh, Tan Lai Ngoc, Tan Minh Nguyen.
International Conference on Machine Learning (ICML), 2026.
5. **Geometric and Stochastic Analysis of Discontinuities in Sparse Mixture-of-Experts**
Tho Tran Huu[†], Huu-Tuan Nguyen, Thien-Hai Nguyen, Nhat-Tri Ho, **Viet-Hoang Tran**, Tho Quan, Tan Minh Nguyen.
International Conference on Machine Learning (ICML), 2026 (spotlight).
6. **Quasi-Equivariant Metanetworks**
Viet-Hoang Tran[†], An Nguyen The[†], Benoit Gruerand, Thieu Vo, Tan Minh Nguyen.
International Conference on Learning Representations (ICLR), 2026.
7. **Mixed-Curvature Tree-Sliced Wasserstein Distance**
Duy-Tung Pham[†], **Viet-Hoang Tran**[†], Thieu Vo, Tan Minh Nguyen.
International Conference on Learning Representations (ICLR), 2026.
8. **Tree-sliced Sobolev IPM**
Viet-Hoang Tran[†], Thanh Tran[†], Thanh Chu, Duy-Tung Pham, Trung-Khang Tran, Tam Le, Tan Minh Nguyen.
International Conference on Learning Representations (ICLR), 2026.

9. **Revisiting Tree-Sliced Wasserstein Distance Through the Lens of the Fermat-Weber Problem**
Viet-Hoang Tran[†], Thanh Tran[†], Thanh Chu[†], Duy-Tung Pham, Trung-Khang Tran, Tam Le, Tan Minh Nguyen.
International Conference on Learning Representations (ICLR), 2026.
10. **On Linear Mode Connectivity of Mixture-of-Experts Architectures**
Viet-Hoang Tran[†], Van Hoan Trinh[†], Khanh Vinh Bui[†], Tan M. Nguyen.
Advances in Neural Information Processing Systems (NeurIPS), 2025 (oral).
11. **Tree-Sliced Entropy Partial Transport**
Viet-Hoang Tran[†], Thanh Tran, Thanh Chu, Tam Le, Tan Minh Nguyen.
Advances in Neural Information Processing Systems (NeurIPS), 2025.
12. **Dynamical Properties of Tokens in Self-Attention and Effects of Positional Encoding**
Duy-Tung Pham[†], An Nguyen The[†], **Viet-Hoang Tran**, Nhan-Phu Chung, Xin T. Tong, Tan Minh Nguyen, Thieu Vo.
Advances in Neural Information Processing Systems (NeurIPS), 2025.
13. **Tree-Sliced Wasserstein Distance with Nonlinear Projection**
Thanh Tran[†], **Viet-Hoang Tran**[†], Thanh Chu, Trang Pham, Laurent El Ghaoui, Tam Le, Tan M. Nguyen.
International Conference on Machine Learning (ICML), 2025.
14. **Tree-Sliced Wasserstein Distance: A Geometric Perspective**
Viet-Hoang Tran[†], Trang Pham[†], Tho Tran, Minh Khoi Nguyen Nhat, Thanh Chu, Tam Le, Tan M. Nguyen.
International Conference on Machine Learning (ICML), 2025.
15. **Equivariant Polynomial Functional Networks**
Thieu N. Vo[†], **Viet-Hoang Tran**[†], Tho Tran Huu[†], An Nguyen The, Thanh Tran, Minh-Khoi Nguyen-Nhat, Duy-Tung Pham, Tan Minh Nguyen.
International Conference on Machine Learning (ICML), 2025.
16. **Equivariant Neural Functional Networks for Transformers**
Viet-Hoang Tran[†], Thieu N. Vo[†], An Nguyen The[†], Tho Tran Huu, Minh-Khoi Nguyen-Nhat, Thanh Tran, Duy-Tung Pham, Tan Minh Nguyen.
International Conference on Learning Representations (ICLR), 2025.
17. **Distance-Based Tree-Sliced Wasserstein Distance**
Viet-Hoang Tran[†], Khoi N.M. Nguyen[†], Trang Pham, Thanh T. Chu, Tam Le, Tan M. Nguyen.
International Conference on Learning Representations (ICLR), 2025.
18. **Spherical Tree-Sliced Wasserstein Distance**
Viet-Hoang Tran[†], Thanh T. Chu[†], Khoi N.M. Nguyen, Trang Pham, Tam Le, Tan M. Nguyen.
International Conference on Learning Representations (ICLR), 2025.
19. **Monomial Matrix Group Equivariant Neural Functional Networks**
Viet-Hoang Tran[†], Thieu N. Vo[†], Tho Tran-Huu, An T. Nguyen, Tan M. Nguyen.
Advances in Neural Information Processing Systems (NeurIPS), 2024.

Workshop Publications

1. **Modeling Expert Interactions in Sparse Mixture of Experts via Graph Structures**
Minh-Khoi Nguyen-Nhat[†], Rachel S.Y. Teo, Laziz Abdullaev, Maurice Mok, **Viet-Hoang Tran**, Tan Minh Nguyen.
AAAI 2026 Workshop MATH4AI.
2. **Equivariant Neural Functional Networks for Transformers**
Viet-Hoang Tran[†], Thieu N. Vo[†], An Nguyen The[†], Tho Tran Huu, Minh-Khoi Nguyen-Nhat, Thanh Tran, Duy-Tung Pham, Tan Minh Nguyen.
ICLR 2025 Workshop Weight Space Learning (spotlight).
3. **Distance-Based Tree-Sliced Wasserstein Distance**
Viet-Hoang Tran[†], Khoi N.M. Nguyen[†], Trang Pham, Thanh T. Chu, Tam Le, Tan M. Nguyen.
ICLR 2025 DeLTa Workshop.

4. Spherical Tree-Sliced Wasserstein Distance

Viet-Hoang Tran[†], Thanh T. Chu[†], Khoi N.M. Nguyen, Trang Pham, Tam Le, Tan M. Nguyen.
ICLR 2025 DeLTa Workshop.

Preprints

1. A Clifford algebraic approach to E(n)-equivariant high-order graph neural networks

Viet-Hoang Tran[†], Thieu N. Vo[†], Tho Tran, Tan Nguyen.
preprint.

HONORS AND AWARDS

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| • Graduate Tutor Merit Award (Tier 1) | 2026 |
| • ICML 2026 Gold Reviewer | 2026 |
| • Graduate Tutor Merit Award (Tier 1) | 2025 |
| • NUSGS Research Incentive Award | 2023 |
| • NUS President's Graduate Fellowship | 2022 |
| • International Mathematics Competition for University Students, <i>Grand First Prize</i> | 2020 |
| • Vietnam Mathematical Olympiad for University Students, <i>First Prizes in Algebra and Analysis</i> | 2019 |
| • International Mathematical Olympiad, <i>Bronze Medal</i> | 2018 |
| • Vietnam Mathematical Olympiad for High School Students, <i>First Prize</i> | 2018 |

TEACHING EXPERIENCE

National University of Singapore

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|---|-------------|
| MA2104 Multivariable Calculus | Spring 2026 |
| MA2104 Multivariable Calculus | Fall 2025 |
| DSA2102 Essential Data Analytics Tools: Numerical Computation | Spring 2025 |
| MA2311 Techniques in Advanced Calculus | Fall 2024 |

Tran Phu High School for Gifted Students

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| Mathematics for Vietnam Mathematical Olympiad | Fall 2018–Spring 2023 |
| Mathematics for Vietnam IMO Team Selection Test | Fall 2018–Fall 2022 |

TALKS AND ACADEMIC ACTIVITIES

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| • Talk: <i>Symmetry in Parameter Space</i>
Workshop on Mathematical Approaches to Modern AI Challenges, Institute for Mathematical Sciences, National University of Singapore | Sep 2026 |
| • Talk: <i>Symmetry in Parameter Space</i>
Workshop on Mathematical Foundations of AI Models, Institute for Mathematical Sciences, National University of Singapore | Aug 2026 |
| • Poster presentation
International Conference on Machine Learning 2026, Seoul, South Korea | Jul 2026 |
| • Talk: <i>Functional Equivalence in Neural Networks</i>
Weekly Seminar, Department of Mathematics, University of California, Los Angeles | Apr 2026 |
| • Poster presentation
International Conference on Learning Representations 2026, Rio de Janeiro | Apr 2026 |
| • Talk: <i>Functional Equivalence in Neural Networks</i>
Hanoi–Singapore Meeting, Vietnam Institute for Advanced Study in Mathematics, Hanoi | Apr 2026 |

- Talk: *Symmetry in Parameter Space* Feb 2026
Graduation Day, Information Theory and Applications Workshop 2026, San Diego
- Talk: *Symmetry in Parameter Space: Theory and Application* Dec 2025
Machine Learning Seminar, SOICT-HUST, Hanoi University of Science and Technology
- Oral presentation: *On Linear Mode Connectivity of Mixture-of-Experts Architectures* Dec 2025
Poster presentation
Conference on Neural Information Processing Systems 2025, San Diego
- Talk: *Functional Equivalence in Neural Architectures* Jul 2025
Mathematical Seminar, Faculty of Mathematics, Hanoi University of Science
- Poster presentation Jul 2025
International Conference on Machine Learning 2025, Vancouver
- Talk: *Functional Equivalence in Neural Architectures* Jun 2025
Joint FDU-NUS-ZJU PhD Forum in Mathematics, National University of Singapore
- Talk: *Equivariant Neural Functional Networks for Transformers* Apr 2025
ICLR 2025 Workshop on Weight Space Learning, Singapore
- Poster presentation Apr 2025
International Conference on Learning Representations 2025, Singapore
- Information Theory and Applications Workshop 2025, San Diego Feb 2025
- Poster presentation Dec 2024
Conference on Neural Information Processing Systems 2024, Vancouver
- Talk: *Algebraic Perspective of Neural Architecture* Oct 2024
Seminar, Department of Electrical and Computer Engineering, Rice University
- SIAM Conference on Mathematics of Data Science 2024, Atlanta Oct 2024
- International Conference on Machine Learning 2024, Vienna Jul 2024
- Poster presentation Mar 2024
Workshop on Optimal Transport, Berlin
- The 62nd IEEE Conference on Decision and Control, Singapore Dec 2023
- Seminars in Algebraic Geometry, Vietnam Institute of Mathematics 2019–2022
- Talk: *Introduction to Elliptic Curves* Aug 2020
Workshop on The Mordell-Weil Theorem, Hanoi
- Boundary Behaviour of Holomorphic Mapping May–Jul 2020
Scientific Research for University Students Program, Vietnam Institute of Mathematics
- SEAMS-School on Arithmetic, Geometry and Model Theory, Hanoi Feb 2020
- IMH-School on Introduction to Algebraic Schemes and Cohomology, Hanoi Nov 2019